Docker & Docker Compose

1. What is Docker?
    1. Docker is an open-source platform designed to run application, automate the deployment and scaling inside a lightweight portable **containers**. Containers includes the application along with its dependencies, ensuring consistent behavior across different systems.

2. Docker Architecture:
    - ➔ Docker Client: It's a CLI tool. It sends the docker commands to docker daemon.
    - ➔ Docker Host: It consist of docker daemon, which manages everything in background such as images, containers.
    - ➔ Docker Registry: It stores each and every docker images that is created. It is similar to GitHub repository which tracks every code change.
    - ➔ Docker Image: It's a blueprint to create a docker container, based on docker image the application is build. It's a read-only template. Docker images are immutable.
    - ➔ Docker Container: It is a live executable instance of a docker image. These are mutable.
    - ➔ Docker Hub: It is a public registry to find, store, and share docker images.
    - ➔ Docker Daemon(Engine): It runs in the background, helps to manage images, containers, networks and volumes.
    - ➔ Worklfow:
        - Users execute a Docker command in cli.
        - Docker Client sends the request to the docker Daemon.
        - Docker Daemon build or downloads the required image.
        - Docker Daemon creates and runs the container.

3. Images vs Containers

Docker Images	**Docker Containers**
A read-only template used to create containers.	A live running instance of a Docker image.
Cannot be modifies while running.	Can be started, stopped, and removed.
Used as a blue-print for applications.	Executes the application inside an isolated environment.
Stored locally or in a registry like Docker Hub.	Exists only while running or until deleted.

4. Dockerfile:
    - ➔ Dockerfile is a plain-text file containing instructions that tells docker how to build an image step-by-step. It defines the base image, application files, dependencies, exposed ports and startup commands.

Example:
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80

FROM: Sets the base image
COPY: Copies the files into the image
EXPOSE: Exposes the port inside a containers

5. Docker Volumes:
   - ➔ Volumes are like external storage lockers attached to containers.
     Used for the persistent storage for containers. Data stored in a volume remains available even if the container is stopped or deleted.
     Also allows the data sharing between multiple containers.
     Why?
   - ➔ Data inside containers is lost on removal, then volumes are used to persist the data beyond the container file.
   - ➔ Example:
     docker volume create website-task

6. Docker Network
   - ➔ Docker containers run in a isolated environment. In real-world scenarios like web application with database, containers often need to communicate and that's where docker networking plays a key role.
   - ➔ There are different types of docker networks:
     1. Bridge Network(Default)
     2. Host Network
     3. None Network

7. Docker Compose
   - ➔ Docker Compose is a tool is used to define and manage multi-container docker applications using the yaml configuration docker-compose.yml file.
   - ➔ Mostly its used for the testing the production environment.
   - ➔ It supports the networking and the volume configuration.
   - ➔ Controls the group of containers via single commands, docker compose up command and docker compose down command.
   - ➔ Example:
     docker compose up -d

8. Docker Hub
   - ➔ Docker Hub is a cloud-based registry used to store, manage, and share docker images.
   - ➔ It has both public and private repositories.

- Also has integration with the CI/CD pipelines.
- Example commands:
  docker login

  docker push username/image-name:version

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